Half-Plane Shielding: Provable Directional COLREGs Compliance for Continuous-Action Safe Reinforcement Learning

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ABSTRACT

Objectives: Autonomous surface vessels must obey the International Regulations for Preventing Collisions at Sea (COLREGs). The give-way clauses are directional. A track that avoids a collision but passes on the wrong side still violates them. This paper establishes a per-step guarantee of directional compliance for continuous control on give-way steps where the projection is feasible.

Methods: The give-way clause is written as a half-plane constraint on the control plane. It is intersected with the actuator limits and a one-step collision-free condition, giving a convex safe control set. A quadratic program with two variables and at most six linear constraints projects the control command of the policy onto the nearest point of that set. The projection sits inside the environment transition, so the policy learns under its corrected commands. The scenario library holds 2,000 simulated two-vessel encounters, 1,400 for training and 600 for test. Nine configurations are compared, each trained with eight seeds.

Findings: On every give-way step where the projection is feasible, directional compliance holds by construction, independently of the training outcome and the seed. Emergency and fallback steps are not covered. On the 600 test scenarios the median violation count is 0.53 per episode at a median collision rate of 0.00%, against 1.16 for the discrete shielded baseline. In the ablation the bounded action distribution accounts for a 65% lower median turn increment (paired signed-rank $p = 0.008$). The arrival rate is lower and is not claimed as a benefit.

Novelty: To the best of the authors' knowledge, this is the first projection-based safety shield for directional COLREGs compliance under continuous control.

Practical Applications: The safety shield is independent of the training outcome. The online cost is one quadratic program per step, with a median solve time of 4.11 ms against a decision period of 10 s.

1 INTRODUCTION

At sea, collision avoidance is coordinated by a shared rule set. Each vessel is expected to act as the rules prescribe, and each can expect the same of the other. When no one is on board, that predictability becomes a technical requirement. A maritime inquiry does not only ask whether contact occurred. It also asks whether the vessel acted as the rules require. An encounter that ends without contact but passes on the wrong side is still recorded as a breach.

Collision avoidance for autonomous surface vessels must therefore satisfy the International Regulations for Preventing Collisions at Sea, or COLREGs (IMO 1972). The give-way clauses are directional. In a head-on encounter both vessels turn to starboard. In a crossing encounter the give-way vessel turns to starboard and the stand-on vessel holds course and speed. In an overtaking encounter the overtaking vessel keeps clear. A control command can therefore avoid contact and still break the rules, because it turned the wrong way. Compliance cannot be verified from the outcome. It is a condition on which half-plane of the control plane each command falls in, and that is not the usual shape of a constraint in safe reinforcement learning.

Learning-based work on directional compliance follows two routes, and the two keep different properties. The first discretizes the action space and masks the non-compliant commands. Compliance then holds by construction, and it does not depend on the training outcome or on the seed. Quantization fixes the control resolution at the grid spacing. A give-way turn that must be readily apparent has a smallest admissible turn rate. On a fixed grid the smallest available compliant turn rate is the nearest grid point at or beyond it. The second route keeps continuous control and writes the rules into the reward as a penalty (Meyer et al. 2020; Heiberg et al. 2022; Müller et al. 2026). The control resolution is preserved. A reward acts on expected behavior, so it excludes no single violating command. Section 2 sets out both routes in detail.

To the best of the authors' knowledge, continuous action spaces, provable directional compliance, and COLREGs have not been combined before. This paper addresses that combination with a projection-based safety shield. A rule state machine returns the half-plane of compliant directions. Intersecting that half-plane with the actuator limits and a one-step collision-free condition gives a convex safe control set. A quadratic program projects the continuous control command of the policy onto the nearest point of that set. Projection replaces enumeration because it needs the safe control set to be convex rather than enumerable. The shield sits inside the environment transition, so the policy learns under its own corrected commands.

The word provable covers different scopes in the safe reinforcement learning literature, and those scopes do not always match what a reader assumes. Each guarantee below is therefore stated with its assumptions, and the cases that are not covered are stated as well.

Contributions.

1. The half-plane form of the give-way clause (Section 4.1). Under the action-space formulation adopted here, the directional requirement is a single half-plane on the plane of control commands. Intersected with the actuator limits and a one-step collision-free condition it gives a convex safe control set, so the premise that the set must be enumerated can be replaced by the premise that it is convex.

2. A projection-based COLREGs shield for continuous action spaces (Section 4.2). A two-variable quadratic program projects the policy command onto that set inside the environment transition. Directional compliance holds by construction on every give-way step where the projection is feasible, independently of the training outcome and of the seed. Emergency steps and steps with an empty safe control set are excluded.

3. Three scoped propositions (Sections 4.2.1 to 4.2.3): a sufficient far-field condition for one-step collision freedom, the compliance statement above, and a one-sided criterion for an unavoidable collision. Each carries its assumptions.

4. Measured effect on the official CommonOcean split (Section 5). Over eight seeds per configuration, the full method records 0.53 COLREGs violations per episode against 1.16 for the discrete shielded baseline and 4.49 for the control barrier function at its lowest-violation setting. A nine-configuration ladder separates the effect of each mechanism.

2 RELATED WORK

2.1 Maritime Collision Avoidance and COLREGs Compliance

Work on rule compliance began with model-driven methods. The velocity obstacle selects an avoiding velocity from the relative velocity cone. (Kuwata et al. 2014) gave it COLREGs semantics, and (Huang et al. 2019) extended it to more general encounters. Model predictive control scores candidate maneuvers over a finite horizon on both risk and compliance (Johansen et al. 2016). Its branching-course variant has been tested at sea (Eriksen et al. 2019) and at full scale (Kufoalor et al. 2020). An earlier protocol-based method wrote the rules as an agreement negotiated between vessels (Benjamin et al. 2006). Recent work on this route uses control barrier functions and convex optimization (Lee et al. 2026; Patil et al. 2026), mostly on standard maneuvering models (Fossen 2011). Their guarantees come from explicit constraints. The control law is designed by hand and contains no learned policy.

Deep reinforcement learning learns the policy from data (Sarhadi et al. 2022). Most of that work discretizes the control command. (Shen et al. 2019) use a finite set of rudder angles for confined water. (Zhao and Roh 2019; Chun et al. 2021) discretize avoidance timing and paths instead. Discretization helps optimization and fixes control resolution at the grid spacing. A second group keeps continuous control. (Meyer et al. 2020) write COLREGs as a soft penalty that decays with bearing and range, and (Heiberg et al. 2022) write them as a tunable risk gate. Compact reinforcement learning for obstacle avoidance has also been shown on underactuated vessels (Cheng and Zhang 2018). This group is usually built on proximal policy optimization (Schulman et al. 2017). A bounded Beta distribution (Chou et al. 2017) can parameterize the continuous control command. Its samples are then not clipped at the boundary of the action box. Control resolution is preserved. Rule compliance is induced by the reward, so no single violating command is excluded.

2.2 Safe Reinforcement Learning and Action Shielding

Safe reinforcement learning (García and Fernández 2015) attaches hard guarantees to a learned policy. Constrained Markov decision processes state safety as an expected cumulative cost (Altman 1999). A constraint of that form is satisfied in expectation, so it again excludes no single control command. Action masking removes unsafe commands from a discrete action space according to a temporal logic specification (Alshiekh et al. 2018). A continuous-space form of it has been developed for general tasks, without maritime rules (Kim et al. 2024). In continuous control, quadratic programs built on control barrier functions (Ames et al. 2019) and predictive safety shields (Wabersich and Zeilinger 2021) give per-step hard guarantees. The latter has been attached to maritime reinforcement learning, where it constrains collision only and carries no directional clause (Vaaler et al. 2024). (Dalal et al. 2018) linearize a state constraint at the current state and project the control command onto the resulting half-space in closed form. In the present formulation the feasible set combines the actuator limits, a directional COLREGs constraint, and a one-step collision-free constraint. The projected quantity is therefore a rule-compliant control command rather than a state bound.

Table 1 places this work on four dimensions. Among the maritime studies listed there, a per-step hard guarantee of directional rule compliance appears only on a discrete action space (Krasowski and Althoff 2024). To the best of the authors' knowledge that also holds more broadly. That work also constructed the scenario set used here, inside the CommonOcean benchmark framework (Krasowski and Althoff 2022), and reports the state machine thresholds adopted here. It identifies action correction under continuous control as future work, and names action projection (Kochdumper et al. 2023) as one mechanism. Later work on that direction moved to continuous control but approximated compliance with a soft reward, without a hard guarantee (Müller et al. 2026). The present work gives a constructive guarantee of directional compliance under continuous control. (Markgraf et al. 2026) compare projection placed inside the environment with projection placed inside the policy, and analyze how each affects the policy gradient. Here the safe control set is a convex region of the control plane and the correction is a quadratic program in two variables, which keeps the per-step cost small.

A directional constraint tightens the feasible set in close quarters and can cost terminal efficiency. Potential-based shaping recovers it without changing the set of optimal policies (Ng et al. 1999), but a distance potential alone leaves a stalling mode near the goal (Trott et al. 2019; Müller and Kudenko 2025). Shaping can instead be designed against a stated control requirement (De Lellis et al. 2024). Here a linear distance kernel is used, with arrival as a terminal condition. Applying the safety shield during training and penalizing the correction improves sample efficiency (Pizarro Bejarano et al. 2025). A sufficiently permissive shield does not cost asymptotic optimality (Oh et al. 2025). A safety term that opposes the goal term causes stalling short of the goal (Grover et al. 2023).

TABLE 1 Position of related work. A per-step hard guarantee means the class of control commands the method can exclude at each *covered* decision step, not constraint satisfaction in expectation. For Ours the covered steps are the give-way steps on which the projection is feasible; emergency and fallback steps are excluded.

Representative work	Action space	Compliance is imposed by	Per-step guarantee	Learned
Model-driven, no learning				
(Kuwata et al. 2014)	Continuous	Velocity obstacle geometry	Geometric	-
(Johansen et al. 2016)	Branches	Cost ranking	-	-
(Lee et al. 2026; Patil et al. 2026)	Continuous	Barrier function	State bound	-
Learning, compliance from a soft reward				
(Meyer et al. 2020)	Continuous	Soft reward	-	✓
(Heiberg et al. 2022)	Continuous	Soft reward, risk gate	-	✓
(Müller et al. 2026)	Continuous	Soft reward, falsification	-	✓
Learning, compliance from a hard constraint				
(Alshiekh et al. 2018)	Discrete	Action masking	State bound	✓
(Dalal et al. 2018)	Continuous	Linearize, then project	State bound	✓
(Vaaler et al. 2024)	Continuous	Predictive safety shield	Collision only	✓
(Krasowski and Althoff 2024)	Discrete (49)	Action masking	**Direction**	✓
Ours	**Continuous**	**Projection (quadratic program)**	**Direction**	✓

3 PROBLEM FORMULATION

3.1 Vessel Kinematics

The controlled vessel is modeled as a point mass with a yaw constraint. Its state is $s = [p_x, p_y, \theta, v]^\top \in \mathbb{R}^4$, which collects position, heading, and speed. Its controls are the longitudinal acceleration a and the turn rate ω,

$$\dot{s} = \left[v\cos\theta,\;\; v\sin\theta,\;\; \omega,\;\; a \right]^\top, \qquad u = [a, \omega]^\top. \tag{1}$$

The controls obey $|a| \le a_{\max}$ and $|\omega| \le \omega_{\max}$, and the speed obeys $0 \le v \le v_{\max}$. A command is held constant over one decision period Δt. Headings and bearings are wrapped to $(-\pi, \pi]$. A relative bearing is positive to starboard. The turn rate follows the usual mathematical convention and is positive counterclockwise, so a turn to starboard has $\omega < 0$.

The speed bound is applied after integration, so the speed can briefly exceed $v_{\max}$ within a step. Every one-step displacement bound in this paper therefore uses $v_{\max}\Delta t + \frac{1}{2}a_{\max}\Delta t^2$, and the speed at the end of a step is at most $v_{\mathrm{bnd}} = v_{\max} + a_{\max}\Delta t$.

This level of detail is chosen because the give-way rules govern changes of course and of speed, and Equation (1) takes both as control commands. The rules can therefore be written directly as constraints on u, which is what lets the safety shield live in the action space. A finer hydrodynamic model would add coefficients that differ by vessel and are hard to identify. Wind, current, and waves are not modeled.

The own vessel is a large container vessel of length 175 m and beam 25.4 m. Its limits are $a_{\max} = 0.24$ m/s^2, $\omega_{\max} = 0.03$ rad/s, and $v_{\max} = 9.5$ m/s, and the decision period is 10 s. The vessel model, these limits and the decision period are those reported by (Krasowski and Althoff 2024). The circumscribed radius of the hull is 88.4 m and the inscribed radius is 12.7 m, both computed from the stated rectangular hull dimensions.

The turn rate limit is about 1.7° per second. A readily apparent turn of 20° therefore needs more than ten seconds, which is longer than one decision period. The maneuver time therefore exceeds one decision period, so the safety constraint has to look ahead.

The hull is a rectangle. Its circumscribed circle over-approximates it and supports conditions of the form “contact is impossible”, while its inscribed circle under-approximates it and supports conditions of the form “contact is certain”. One circle in each direction keeps both kinds of conclusion one-sided. The target vessel is a rectangle as well, extrapolated at constant velocity whenever a prediction is needed.

3.2 Observation and Action Spaces

The observation is a real vector of 27 entries in four groups. They are four for the state of the own vessel, six for the relation to the goal, twelve for the traffic situation, and five termination flags. The traffic group uses four bearing sectors, each contributing the nearest target vessel in it with its distance, its relative bearing, and the change of that distance over one step. The observation layer outputs raw physical quantities and the training wrapper normalizes them. Organizing traffic by sector keeps the observation dimension fixed. The shield is implemented for a single target vessel, and it raises an error rather than running in a degraded mode when more are present.

The action is the two-dimensional command $u = (a, \omega)$, and two boxes have to be distinguished. The policy acts in the narrower operating box, whose half-widths are 0.048 m/s^2 and 0.018 rad/s. These reproduce the span of the 7×7 discrete action grid reported by (Krasowski and Althoff 2024). The two action spaces therefore carry the same control authority and differ only in resolution. The shield and the emergency controller act in the wider box of the actuator limits, $U_{\text{box}} = \{|a| \le a_{\max}\} \cap \{|\omega| \le \omega_{\max}\}$.

The widening is necessary. A shield confined to the narrow box would declare a step infeasible whenever that box and the collision-free constraint disagree, even though the step is in fact solvable. The cost is that the applied command can fall outside the policy action box, which is returned in the observation.

3.3 Compliance as a Per-Step Constraint

We consider a two-vessel encounter in open water under five assumptions. (A1) Open water, with no traffic separation scheme, no static obstacle, and no shoreline constraint. (A2) One controlled vessel and one target vessel, both power-driven. (A3) The dynamics of the controlled vessel are Equation (1), applied with zero-order hold at period Δt. (A4) The current state of the target vessel is available without measurement error. (A5) No give-way obligation is already in force at the initial instant.

Assumption (A2) applies the pairwise obligations of the convention (IMO 1972) to one controlled vessel and one target vessel. Three or more vessels can meet at once, and their obligations can then conflict. One case is an obligation to hold course for one vessel and to give way to another. The convention does not say which obligation prevails, and this paper does not define a precedence rule.

Let ρ_t be the encounter situation at time t. Let $U_{\text{colregs}}(\rho_t) \subseteq \mathbb{R}^2$ be the set of commands that the rules allow in that situation, and let $U_{\text{cf}}(s_t)$ be the collision-free set. The problem is to find a policy

$$\pi : S \to \mathbb{R}^2 \quad \text{such that} \quad u_t \in U_{\text{colregs}}(\rho_t) \cap U_{\text{cf}}(s_t) \quad \text{on every covered step.} \tag{2}$$

A covered step is a non-emergency step on which the constrained set is non-empty. Emergency steps and steps with an empty set are handled separately in Section 4.2.3, and they are excluded from every guarantee in this paper. Equation (2) is not the same as writing compliance into the reward. It constrains the command that is executed rather than an expected value.

When the action set is finite, Equation (2) has a direct solution: enumerate every action, test it, and mask out what does not comply. That route operates on an enumerable set of commands. Rudder and propulsion are continuous, so quantizing them fixes the available control levels and their spacing, and a different enforcement operator is needed. Enumeration is therefore replaced by projection. The give-way clause is a half-plane constraint on the command (Section 4.1). The collision-free condition can be written as a linear inequality (Section 4.2). The feasible set of Equation (2) is therefore the intersection of a rectangle with a few half-planes in the plane, which is a convex polygon. The projection onto a convex set exists and is unique, and it is obtained from a very small quadratic program.

4 METHODOLOGY

4.1 COLREGs Obligations as Half-Plane Constraints

The rule state machine implements Rules 13 to 17 and the stand-on obligation to hold course and speed, and sorts the encounter into six situations,

$$\rho \in \{\rho_0 \text{ conflict-free},\ \rho_1 \text{ stand-on},\ \rho_2 \text{ head-on},\ \rho_3 \text{ crossing},\ \rho_4 \text{ overtaking},\ \rho_5 \text{ emergency}\},$$

The decision is a conjunction of three groups of conditions. Each condition is a predicate, that is, a named test on the two states that returns true or false.

The first group is the bearing sector, taken along the sidelight boundaries of the convention. The second group is the heading relation, which separates nearly anti-parallel from nearly parallel headings. The third group is collision possibility. It has two parts. The speed set of the own vessel must intersect the collision cone of the target vessel, that is, the set of relative velocities that lead to contact. The relative motion must also reduce the current center distance within the check horizon. The forward sector has a half-angle of 5° and the side sectors end at 112.5°. The overtaking band has a half-angle of 67.5°. The collision-cone radius is three times the length of the target vessel, as reported by

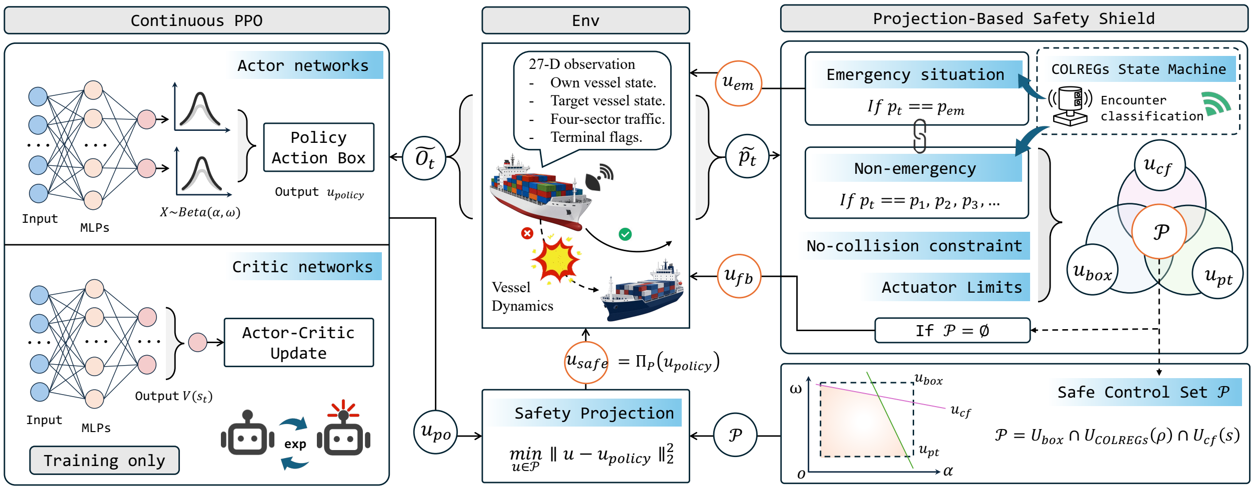

Figure 1 Overview of the method. The safe control set U_{safe} is the intersection of the actuator limits, the half-plane of compliant directions, and the one-step collision-free constraint. The projection is part of the environment transition, so the shield sits inside the training loop. The two bypasses are exclusive. MLP denotes a multilayer perceptron.

(Krasowski and Althoff 2024). This detects an approach with less than about two target-vessel lengths of clearance. The check horizon is 420 s.

Four situations carry an obligation. More than one of their predicates can hold at the same instant, so the four are tested in a fixed priority order and the first match wins. The rows below are written in that priority order, which is not the order of the indices,

$$\begin{aligned} \rho_2 \text{ head-on}: &\quad \mathrm{CP} \wedge \mathrm{Fr}(s_e, s_m) \wedge \mathrm{Ap}, \\ \rho_3 \text{ crossing}: &\quad \mathrm{CP} \wedge \mathrm{Ri}(s_e, s_m) \wedge \mathrm{Tl}, \\ \rho_4 \text{ overtaking}: &\quad \mathrm{CP} \wedge \mathrm{Be}(s_m, s_e) \wedge (v_e > v_m) \wedge \mathrm{Sd}, \\ \rho_1 \text{ stand-on}: &\quad \big[\mathrm{CP} \wedge \mathrm{Le}(s_e, s_m) \wedge \mathrm{Tr}\big] \;\vee\; \rho_4(s_m, s_e), \end{aligned} \tag{3}$$

and all other cases fall into ρ_0. Stand-on is tested last, and its definition reuses the overtaking predicate with the two vessels exchanged, because overtaking places the own vessel in the aft sector of the target vessel.

Entry into a give-way situation has two routes. Under the persistence route the predicate must not hold now, and it must hold at every decision step of a 60 s reaction window under constant-velocity extrapolation. Under the immediate route the obligation is entered when the predicate already holds at the current step. The persistence route alone distinguishes an obligation that is about to apply from one that already applies, which leaves the asymmetry that Section 4.3.3 removes. The emergency situation is entered by a stronger set-based prediction with a horizon of 180 s.

Each situation maps to an explicit constraint set on the command,

$$U_{\text{colregs}}(\rho) = \begin{cases} U_{\text{box}}, & \rho = \rho_0, \\ \{|a| \le \varepsilon_a\} \cap \{|\omega| \le \varepsilon_\omega\}, & \rho = \rho_1, \\ \{\omega \le -\omega_{\text{turn}}\}, & \rho \in \{\rho_2, \rho_3\}, \\ \{\sigma\, \omega \le -\omega_{\text{turn}}\}, & \rho = \rho_4, \\ U_{\text{box}}, & \rho = \rho_5. \end{cases} \tag{4}$$

Head-on and crossing require a turn to starboard. The convention does not fix the side for overtaking. The state machine selects it from the relative heading of the two vessels and returns the sign $\sigma \in \{+1, -1\}$. Only the selected side is allowed on that step. Stand-on holds course and speed. Each set is therefore a single half-plane or an interval on the plane of control commands, and none of them is a union.

The convention asks a give-way action to be large enough to be seen in time. We quantify this as a course change of at least a given threshold within the maneuver window. With a window of 40 s and a threshold of 20°,

$$\omega_{\text{turn}} = \frac{\Delta_{\text{lt}}}{T_M} = \frac{20^\circ}{40\ \text{s}} \approx 0.008727\ \text{rad/s}. \tag{5}$$

Under the adopted 20°-within-40 s definition this is the smallest turn rate that meets the threshold.

Encounter classification uses the temporal-logic predicates and the numerical thresholds reported by (Krasowski and Althoff 2024). The present formulation states the collision-cone radius and the half-width of the speed set explicitly. It makes entry into a give-way situation symmetric (Section 4.3.3). It then maps each classified situation to a constraint set on continuous control commands.

4.2 Safe Control Set and Projection

4.2.1 One-Step Collision-Free Constraint

At each decision step the applied command must keep the two hulls apart at the end of the next step. Written directly, that condition is not convex. It is replaced by a convex surrogate constraint, built as follows.

The occupancy of the target vessel after one step is over-approximated by a disk. The center follows a constant-velocity extrapolation, and the radius includes a safety margin. The own vessel is over-approximated by its circumscribed circle. Whenever the center distance is at least the sum of the two radii, written d_{safe}.

Let n be the unit vector from the target vessel to the nominal end point of the own vessel, and let δ be that distance. The end point follows from the policy command clipped to the action box.

The end point is then expanded to first order in the command, $p_e(u) \approx p_{\text{nom}} + J(u - u_{\text{nom}})$. The end point must lie on the safe side of the separating line. This gives one linear inequality in the command:

$$\underbrace{-n^\top J}_{g^\top}\, u \;\le\; \underbrace{-\big(d_{\text{safe}} - \delta + n^\top J\, u_{\text{nom}}\big)}_{h}, \qquad U_{\text{cf}}(s) = \{u : g^\top u \le h\}. \tag{6}$$

A first-order expansion is neither an inner nor an outer approximation. The sign of the remainder changes with the curvature. Equation (6) alone therefore does not establish conservatism. The margin inside the safety distance provides it. With that margin the constraint acts at a center distance of 590–840 m, with a median of 714 m over this library. Hull contact needs a much smaller distance. The margin absorbs the error introduced by the linear approximation. The cost is a tighter constraint that rejects some commands which are in fact safe. Strict one-step collision freedom is proved only under the far-field condition of Proposition 1.

The linearization holds only locally. In the far field a sufficient condition is available that does not rely on it.

Proposition 1 (One-step collision freedom in the far field)**.** *Let d be the center distance between the own vessel and a single target vessel. If*

$$d \;\ge\; d_{\text{safe}} + v_{\text{obs,max}}\Delta t + \rho_{\text{ego}}, \tag{7}$$

then the two conservative circumscribed occupancies do not intersect after one decision step. Here d_{safe} is the sum of the two circumscribed radii, and ρ_{ego} is the largest displacement of the own vessel in one step.

Assumptions. A single target vessel. The speed of the target vessel is at most 9.5 m/s. That bound is not implied by the dynamics and is stated separately. The largest value measured over the 2,000 scenarios of the library is 7.10 m/s. The target vessel holds its speed within the step.

Proof sketch. By the triangle inequality, the center distance after the step is at least d minus the one-step displacement bound of each vessel. It is therefore at least d_{safe}.

Scope. The triangle inequality does not depend on direction, so a head-on encounter is the worst case. With the constants of this library the threshold is about 764 m. The statement covers one step. It does not extend to the steps after it, and it does not cover the near field.

4.2.2 Projection as a Quadratic Program

The safe control set is the intersection of three sets,

$$U_{\text{safe}}(s,\rho) = U_{\text{box}} \cap U_{\text{colregs}}(\rho) \cap U_{\text{cf}}(s), \tag{8}$$

and it is a convex polygon in $\mathbb{R}^2$. The applied command is the Euclidean projection of the policy command onto that set,

$$u_{\text{safe}} = \Pi_{U_{\text{safe}}}(u_{\text{policy}}) = \arg\min_{u \in U_{\text{safe}}} \|u - u_{\text{policy}}\|_2^2. \tag{9}$$

The feasible set is an intersection of half-planes, so it can be written as $U_{\text{safe}} = \{u : Au \le b\}$. The first four rows are the actuator limits. The fifth is the compliant-direction half-plane of Equation (4), which is a single half-plane in every situation. The sixth is the collision-free constraint of Equation (6). With two variables and at most six constraints, the program is small enough to solve online at every decision step. Algorithm 1 states one whole decision step, including the two branches that carry no guarantee.

Algorithm 1 One shielded decision step

Require: own state s_e, target state s_m, policy command u_{policy}
Ensure: applied command u_{applied}
1: $\rho \leftarrow \textsc{Situation}(s_e, s_m)$ ▷ one of six, Equation (3)
2: **if** $\rho = \rho_5$ **then** ▷ emergency, carries no guarantee
3: **return** $\textsc{EmergencyController}(s_e, s_m)$
4: $U_{\text{colregs}} \leftarrow$ compliant set for ρ ▷ Equation (4); a single half-plane or an interval
5: $u_{\text{nom}} \leftarrow \Pi_{U_{\text{box}} \cap U_{\text{colregs}}}(u_{\text{policy}})$ ▷ linearization point
6: $(g, h) \leftarrow \textsc{CollisionFree}(s_e, s_m, u_{\text{nom}})$ ▷ Equation (6)
7: $U_{\text{safe}} \leftarrow U_{\text{box}} \cap U_{\text{colregs}} \cap \{u : g^\top u \leq h\}$ ▷ a convex polygon
8: **if** $U_{\text{safe}} = \emptyset$ **then** ▷ fallback, carries no guarantee
9: **return** $\textsc{Fallback}(s_e, s_m)$
10: **return** $\arg\min_{u \in U_{\text{safe}}} \|u - u_{\text{policy}}\|_2^2$ ▷ Equation (9), two variables

Proposition 2 (Directional compliance on every give-way step)**.** *Consider any decision step that the rule state machine assigns to a give-way situation and on which the projection is feasible. The applied command u_{safe} then lies in the compliant-direction half-plane.*

Assumptions. A single target vessel, and a correct decision by the state machine. The second assumption inherits one conservative bias of the collision-possibility predicate. That predicate uses the center distance, and it can miss a trailing encounter in which the two speeds are close.
Proof sketch. The compliant half-plane enters the quadratic program as a hard constraint. Any feasible point satisfies it, and the projection returns a feasible point. The statement holds by construction. It does not depend on the training outcome or on the random seed.
Scope. Emergency situations are not covered, because their branch bypasses the compliance constraint by design. Steps on which the safe control set is empty and the fallback is used are not covered either.

4.2.3 Fallback and the Unavoidable-Collision Criterion

Lines 2 and 11 of Algorithm 1 are the two branches that carry no guarantee. They are mutually exclusive.
Emergency branch. A stateful geometric controller supplies the command, and its lifetime is one emergency event. It has three modes. A near head-on approach ahead tracks a lateral point fixed at the onset. A target vessel astern that acceleration alone can clear takes a straight-line escape inside the reaction window. Every other case tracks a point that moves with the target vessel. One position-tracking law converts each mode into a control command.
Fallback. If the linearization gives no usable separation direction, the command is projected onto the actuator limits alone. Otherwise the direction constraint is relaxed, and if that still leaves no feasible command the fallback minimizes the collision risk. Both branches solve on the actuator limits, so the turn rate can leave the policy action box and saturation is reported against each box separately.

These branches almost never fire on this scenario library. With course and speed held, the median closest center distance is about 1,300 m, so the safe control set is rarely empty. Purpose-built conflict cases were used during development to exercise the branches; they are synthetic and are not part of the reported results.

Neither the emergency branch nor the fallback offers a guarantee. The emergency branch can still decide whether the situation was already beyond recovery.

Proposition 3 (Conservative criterion for an unavoidable collision)**.** *Consider a single target vessel at constant velocity. Let $R_{\text{box}}(t)$ be an over-approximation of the reachable centers of the own vessel, and let $O(t)$ be the hull rectangle of the target vessel. If*

$$\exists t^* \in [0, T] : \quad R_{\text{box}}(t^*) \subseteq \big(O(t^*) \oplus \text{disk}(r_{\text{insc}})\big), \tag{10}$$

then the collision is unavoidable, and every feasible control sequence meets that target vessel at t^*.

Assumptions. A single target vessel at constant velocity. The hull size of the target vessel must not be overstated, since a beam larger than the true value makes the test unreliable. The horizon is 120 s, evaluated at 241 equally spaced instants.

Proof sketch. The rate of change of the velocity vector is bounded. A double integration gives the two semi-axes of R_{box}, which therefore over-approximates the true reachable centers. The inscribed circle under-approximates the hull. Once every reachable center lies inside the inflated occupancy of the target vessel, no control command can avoid contact.

Scope. The criterion is one-sided. It can miss a case, and it never raises a false alarm. A negative result therefore does not mean that the collision is avoidable. The criterion also acts late rather than early, and maneuvering target vessels are not covered. On 20,000 random scenarios, 735 selected scenarios, and 1,500 randomized stress-test cases it produced no false positive.

4.3 Shielded Policy Optimization

The projection of Equation (9) is placed inside the environment transition. The policy emits u_{policy}, the environment applies u_{safe}, and it returns the successor state and the reward. The policy therefore faces the composed transition

$$s_{t+1} \sim P\big(\cdot \mid s_t,\, \Pi_{U_{\text{safe}}(s_t,\rho_t)}(u_t)\big), \qquad u_t \sim \pi(\cdot \mid s_t), \tag{11}$$

rather than the $P(\cdot \mid s_t, u_t)$ of a correction applied after training. Equation (11) describes the covered steps. On an emergency step the applied command comes from the emergency controller, and on a step with an empty safe control set it comes from the fallback (Section 4.2.3). The policy is trained on the transition it will meet at deployment, so it adapts to the projection during training rather than to a shield introduced afterwards.

All configurations are trained with proximal policy optimization (Schulman et al. 2017). The continuous-action configurations use its standard form and the discrete-action baselines its maskable variant, since masking is what makes a discrete shield possible. Both action spaces therefore share one policy-gradient algorithm. The masking is intrinsic to the discrete method rather than a choice made here.

4.3.1 Reward Function

The reward is a sum of five terms,

$$r = r_{\text{sparse}} + r_{\text{goal}} + r_{\text{velocity}} + r_{\text{deviate}} + r_{\text{colregs}}. \tag{12}$$

A sparse terminal term scores arrival, timeout, stalling, and collision, and charges a fixed penalty for every step under emergency control. A dense approach term rewards the change in distance to the goal over one step. A speed term penalizes speed outside the cruise band [2.5, 8.0] m/s. A cross-track term penalizes lateral deviation from the start-to-goal line and saturates at 2,000 m. A soft compliance term decays with distance. It is active only in the reward-shaping baseline and in the discrete configurations. Every continuous configuration here sets it to zero, because compliance is carried by the hard constraint of Equation (4).

For comparability with the discrete shielded baseline, the reward uses the five-term structure reported by (Krasowski and Althoff 2024). The present implementation differs in three declared respects. The approach term uses the difference of distances between two steps. The soft compliance term is rescaled, because its original calibration sits at a scale about fifty times smaller than these scenarios. In the emergency-release predicate of that work, the printed distance clause reads as an upper bound. The accompanying text describes a distance that is large enough. The present implementation follows the text.

4.3.2 Bounded Action Distribution

A continuous policy usually emits an unbounded Gaussian and clips the sample to the action box. That conflicts with Equation (9) for two reasons, both from the clipping. First, in Stable-Baselines3 2.3.2 the action passed to the environment is clipped, while the action stored for the policy gradient is the unclipped sample. Once the mean leaves the box, moving further out changes no reward, the gradient reaches a plateau, and any penalty on action smoothness loses its grip. Second, the differential entropy of a Gaussian has no upper bound while the entropy coefficient is a constant. The standard deviation can therefore keep growing without improving the reward, which pushes still more samples outside the box.

We use a Beta distribution with support on a closed interval. Actions can then never leave the box, clipping becomes the identity map, both effects are removed, the smoothness penalty is differentiable everywhere, and the entropy is bounded above. The network emits two reals per control axis, and

$$\alpha = \operatorname{softplus}(\tilde{\alpha}) + 1, \quad \beta = \operatorname{softplus}(\tilde{\beta}) + 1, \qquad u = u_{\text{lo}} + (u_{\text{hi}} - u_{\text{lo}})\, z, \quad z \sim \operatorname{Beta}(\alpha, \beta) \tag{13}$$

gives the shape parameters and the applied action. Evaluation uses the mode instead of a sample. The offset of one keeps both shape parameters above unity. Below unity the Beta density is U-shaped and diverges at the two ends, and the mode then sits at an endpoint. Because evaluation always takes the mode, such a policy would return a boundary action at every step, and the aggregate metrics would give no sign of it.

4.3.3 Symmetric Entry into Give-Way Situations

In the state machine, entry into a give-way situation is not symmetric. From the conflict-free situation it is triggered only by the persistence condition. From the stand-on situation a separate immediate condition also applies. We add the symmetric branch. When the persistence condition does not fire, the immediate condition is checked, and the give-way situation is entered if it already holds. The obligation is then recognized at the same moment on both routes. The change alters the command that the shield applies, so training and evaluation must use the same setting.

A purpose-built case shows why the branch is needed. Suppose an encounter already satisfies a give-way predicate at the initial instant. The clause "does not hold now" can then never be satisfied. The persistence condition never fires, and the encounter is never recognized as a give-way situation. On a set of such adversarial cases the persistence route alone gives 0 give-way decision steps. With the immediate branch it gives 201. These cases violate Assumption (A5) by construction. They lie outside the formal scope of this paper and are used only as an implementation diagnostic, so they are reported apart from the main results.

4.3.4 Training Setup

Two settings are chosen here: the entropy coefficient is 0.01 and the action distribution is bounded. Every other hyperparameter uses its default value in the implementation library and is left untuned. The libraries are Stable-Baselines3 2.3.2 (Raffin et al. 2021) and its companion package sb3-contrib 2.3.0, and default refers to the released source of those two versions. The advantage is computed by generalized advantage estimation (Schulman et al. 2016). The discrete and continuous configurations agree on every one of these values, so the hyperparameters are not a source of difference between them. All configurations use running normalization of the observation and the reward. Without it, the large-scale approach term in Equation (12) drove the tested implementation to a degenerate solution that holds station.

5 EXPERIMENTS

5.1 Scenarios and Configurations

The experiments use the 2,000 two-vessel encounters constructed by (Krasowski and Althoff 2024) inside the CommonOcean benchmark framework (Krasowski and Althoff 2022). Each scenario fixes the initial pose and speed of the own vessel. It also fixes the initial state and the planned track of the target vessel. Each scenario defines a goal region with a position gate, a heading gate, and a time limit. The encounter geometry is generated programmatically over a range of relative bearings, crossing angles, and closing speeds. Target vessels are 200 to 300 m long, and an episode lasts at most 170 decision steps, which is 1,700 s.

Classifying the library by the geometric predicates returns crossing and head-on only. The library contains no overtaking scenario, and the official test set holds 395 crossing and 205 head-on cases. The overtaking branch of Equation (4) therefore cannot be tested empirically here.

Of the official split of 1,400 training and 600 test scenarios, the training side is divided again into 1,300 for training and 100 for validation. Validation serves only to monitor training and to select a checkpoint, and the test set is never accessed during training or tuning. The validation and test sets are disjoint.

Table 2 lists nine configurations. They form one connected ladder. Most adjacent comparisons change one declared variable; the three bundled steps listed with the table have to be read jointly.

The discrete configurations are reimplemented here from the published description. Without a shield the continuous configuration can only take the plainest form: an unbounded Gaussian, an asymmetric give-way entry, and no shaping. Giving it the continuous-only shaping would break the symmetry with the discrete baseline. The measured difference between the discrete baseline and this configuration can therefore be negative. The comparison is specific to this baseline rather than a general bound on the effect of shielding.

5.2 Metrics and Evaluation Protocol

Metrics. The main metrics are the arrival rate, the collision rate, the COLREGs violations per episode, and the share of steps under emergency control. The violation count is split into give-way and stand-on violations. Control quality

TABLE 2 The nine configurations

Configuration	Cont.	Shield	Masking	Soft rew.	Shaping	Bounded	Sym. entry
Discrete baseline	×	×	×	×	×	-	-
Discrete, soft reward	×	×	×	✓	×	-	-
Discrete, masking	×	×	✓	✓	×	-	×
Continuous, no shield	✓	×	×	×	×	×	×
Continuous, shield	✓	✓	×	×	×	×	×
Ablation reference	✓	✓	×	×	✓	×	×
Ablation, bounded only	✓	✓	×	×	✓	✓	×
Ablation, symmetric only	✓	✓	×	×	✓	×	✓
Ours	✓	✓	×	×	✓	✓	✓

Notes. A check mark denotes enabled, a cross denotes disabled, and a dash denotes not applicable. Rows run from the discrete baseline to Ours. Three steps of the ladder carry a companion variable that cannot be separated. The action space changes together with the policy distribution, the continuous-only shaping is a bundle of four items, and the bounded distribution also changes the initial exploration scale. The effects of those three steps can only be read jointly with their companion variables.

appears in Table 3, and the behavior of the shield and the single-step solve time in Section 5.6. The reward function differs across configurations, so the episode return is never compared across them.

How violations are counted. Violations do not come from the situation decision of the shield. An offline scorer that sees only the trajectory re-decides at each step, from the bare situation predicates, whether the own vessel is under a stand-on or a give-way obligation. A stand-on violation is counted per step once the signed heading change accumulated over the obligation period exceeds the tolerance, and the accumulator resets when the obligation ends. A give-way violation is counted per encounter, and only when no turn in the required direction reaches the readily apparent threshold. One scorer is applied to every method here, including the geometric controllers that have no state machine.

The counting window and the shield do not coincide. The scorer accumulates the heading change over a whole obligation period, while the shield constrains the change inside a single maneuver window of a different length. Emergency and fallback steps also bypass the constraint by design. The shield therefore does not guarantee a violation count of zero, and this paper does not claim one. Proposition 2 guarantees the turn direction on every covered give-way step. A guaranteed turn direction does not imply a give-way violation count of zero. How often the mismatch occurs was not measured, so the counts below are an empirical trajectory-level measure rather than a direct test of that proposition. Violation counts in the figures are the sum of stand-on and give-way violations.

Training budget. Each configuration is trained with 8 random seeds, one run per seed, and the 72 runs together cover about 730 million environment steps. A separate seed fixes the scenario split, so every run sees the same scenarios, and every configuration inside one table is evaluated on one machine in one pass. Training was spread over several machines running the same software image and code. No bit-level cross-machine check was run, so machine differences are not excluded.

Checkpoint selection. Twenty checkpoints are saved per run at even intervals and each is evaluated once on the validation set. The rule was fixed in advance and has no tunable part. The criterion is the highest arrival rate on the validation set, and ties go to the earlier checkpoint. The test set is never accessed. The validation set holds only 100 scenarios, and the binomial standard error of the arrival rate is about 4 points. Taking the maximum over 20 candidates therefore lifts the figure on that set. On the held-out test set the effect is small. Across the 72 runs the arrival rate at the selected checkpoint exceeds the one at the last checkpoint by a median of 0.2 points. It is lower for 31 of them. The ordering of the configurations and the outcome of the paired tests are the same under both rules.

Statistics. The nine-configuration comparisons use a per-seed paired sign test, whose smallest two-sided value with 8 seeds is $p = 2/2^8 = 0.0078$. The 2×2 ablation comparisons use a per-seed paired Wilcoxon signed-rank test. Error bars are bootstrap 95% intervals over seeds. A run counts as converged when its arrival rate on the validation set reaches 50%; the same runs are selected if the criterion is applied on the test set. Nonconverged runs are counted in the main table as they are, and restricting it to converged runs leaves the ordering unchanged.

TABLE 3 Main comparison, all seeds. Official test set, $N = 600$, at the checkpoint selected on the validation set.

Configuration	n	Conv.	Arrival % [CI]	Coll. %	Viol./ep.	Turn Δ	Thr. Δ	Emerg. %
Discrete baseline	8	**8/8**	97.8 [97, 98]	2.00	3.04	0.0149	0.0160	0.00
Discrete, soft reward	8	7/8	**98.2 [97, 98]**	1.58	2.95	0.0141	0.0169	0.00
Discrete, masking	8	6/8	93.3 [10, 96]	0.08	1.16	0.0112	0.0205	4.64
Continuous, no shield	8	7/8	98.0 [96, 99]	1.25	4.44	0.0267	0.0106	0.00
Continuous, shield	8	5/8	69.0 [16, 91]	0.17	1.02	0.0246	0.0158	2.86
Ablation reference	8	5/8	84.5 [9, 90]	0.33	1.13	0.0140	0.0059	3.36
Ablation, bounded only	8	7/8	90.8 [66, 92]	0.17	0.67	0.0049	0.0076	2.27
Ablation, symmetric only	8	6/8	75.5 [4, 89]	**0.00**	0.82	0.0167	0.0053	2.61
Ours	8	**8/8**	88.2 [84, 90]	**0.00**	**0.53**	0.0054	0.0078	2.42

Notes. The best value in each column is in bold, except in the turn and throttle increment columns, which cannot be compared across action spaces (Section 4.3.4). The share of emergency steps is descriptive and is neither good nor bad.

5.3 Main Comparison

Ours has the lowest violation count per episode of all nine configurations, at 0.53, against 1.16 for the discrete safe baseline and 4.44 for the unshielded continuous configuration. A per-seed paired sign test gives 8 wins out of 8 against every other configuration except the two shielded ablations, at $p = 0.0078$. Ours is also the only shielded configuration that reaches the convergence criterion on all 8 seeds. Its arrival rate is 88.2%, which is below the three unshielded configurations and below the discrete shielded baseline. The arrival rate is reported as measured and is not part of the claim. The isolated shield comparison is the Continuous, no shield versus Continuous, shield pair.

5.4 Ablation Study

The four ablation configurations form a complete 2×2 design over the bounded action distribution and the give-way entry. Everything else is identical, and the comparison is paired by seed. The bounded distribution lowers the median turn increment by 65%, with all 8 seeds moving in the same direction and a paired signed-rank value of $p = 0.008$. It lowers the violations per episode by 41%, again 8 of 8, with $p = 0.008$. The symmetric entry alone raises the turn increment by 19%, with $p = 0.016$, and it lowers the violations per episode by 27%, with $p = 0.008$. With both changes on, the turn increment falls by 61% and the violations by 53%. The combined configuration does not differ significantly from the bounded-only configuration in either turn increment or violations per episode, at $p = 0.74$ and $p = 0.11$. Within this ablation set the bounded action distribution accounts for the observed reduction in turn increments, while both changes are associated with lower measured violation counts. Their effects cannot be separated from the declared companion variable, nor by violation type.

This group is the only place where the smoothness result can be read. All four configurations run the same action-increment penalty, so the difference can be attributed to the bounded action distribution rather than to an extra reward term. A comparison against the discrete baseline would not be valid here, because that penalty applies to a quantity the discrete configurations do not have. Adding the bounded distribution also carries a companion variable, since the initial exploration scale differs by about a factor of 2.5 on the acceleration axis. Restricting this group to converged runs leaves the direction of every comparison unchanged.

5.5 Training Reliability and Sample Efficiency

Figure 3 reports learning speed among the converged runs and the number of runs that meet the convergence criterion.

There is a cost, and it appears in learning speed. On the training scenarios the unshielded discrete baseline reaches an arrival rate of 96% by 2 million steps. The shielded continuous configuration is at 33% at that point, and it reaches 88% at the end of training. The shielded discrete baseline lies between the two, at 51% and 93% (Figure 3b). Panel (c) reports the collision rate on the training rollouts and panel (d) the violation count on the validation set. Among the converged runs the unshielded configurations hold a collision rate near 1.8% throughout training and more than 2.4 violations per episode. The shielded configurations sit at a collision rate of zero and near 0.5 violations per episode. The shielded configurations therefore show lower collision and violation metrics together with slower improvement in arrival rate, which is consistent with a narrower explored region of the action space.

Seed-level convergence differs among the shielded configurations. Under the convergence rule declared in advance, all 8 runs of Ours meet the 50% validation-arrival criterion, and it is the only shielded configuration for which

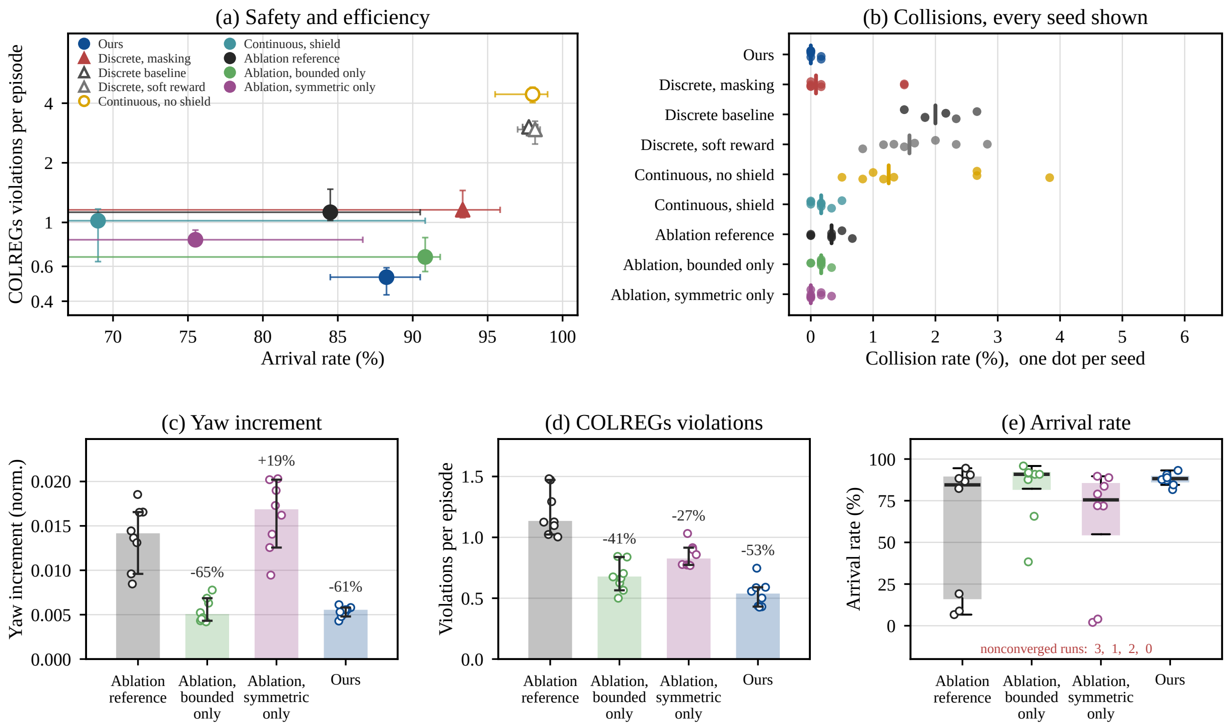

Figure 2 Test-set comparison and ablation. Official test set, the same source as Table 3. Points and bars are medians across seeds, error bars are bootstrap 95% intervals resampled over seeds, and open points are single seeds. The lower right of (a) is the ideal region. Panels (c) to (e) are the 2×2 ablation, paired by seed, with the change relative to the leftmost bar printed above each bar. The arrival rate is bimodal, so (e) is a box plot, and the number of nonconverged runs for the four configurations, in the same left-to-right order, is printed in red at the foot of that panel.

this holds. The shielded ablation configurations reach 5 to 7 seeds, and the discrete safe baseline reaches 6 (Figure 3e). In this eight-seed experiment the two changes therefore increase the number of runs meeting the criterion.

On the test set of 600, 23 of the 48 shielded runs have a collision rate of exactly zero, and 25 do not. The largest is 1.50%, on two seeds of the discrete safe baseline. For Ours, 6 of the 8 seeds have no collision at all and the other two have exactly one collision each, which is 0.17%. A zero count on the 100 validation scenarios cannot establish that the true rate is zero. The 0.00% in Table 3 is the median across eight test-set runs, and two of those runs record one collision each. This agrees with the scope statements of Section 4: a collision rate of zero is an observation and not a guarantee.

5.6 Behavior of the Safety Shield

The emergency branch and the fallback carry no guarantee, so their use is reported separately. At the end of training, the projection produced 93.6% of the control commands. The emergency controller produced 6.2%, and the three fallback branches produced 0.22% in total (Figure 4a). On the test set the share of emergency steps is 2.42%, against 4.64% for the discrete safe baseline. Collisions are rare enough that their source cannot be attributed with confidence. Ours records two collisions in 4,800 test episodes. Neither contradicts Proposition 1 unless its far-field condition held at the step that produced the collision.

The size of the projection correction is normalized by the half-width of the action box. At the end of training its median is 0.082 and its interquartile range is $[0.065, 0.102]$. The correction can be caused by any of the active shield constraints, not only by the compliant-direction half-plane. It also narrows during training (Figure 4c). Panel (b) reports the share of commands that sit against the boundary of the action box. For Ours it ends training near 3.6%, against 98% for the unshielded continuous configuration, which is the effect the bounded action distribution is meant to have. Panel (d) splits the projection steps by situation over the whole of training. Its emergency share of 7.00% is therefore a training-wide average, not the 6.2% measured at the end of training. Among the four shielded continuous configurations the correction is largest for the symmetric-only configuration, at 0.093, and smallest for the

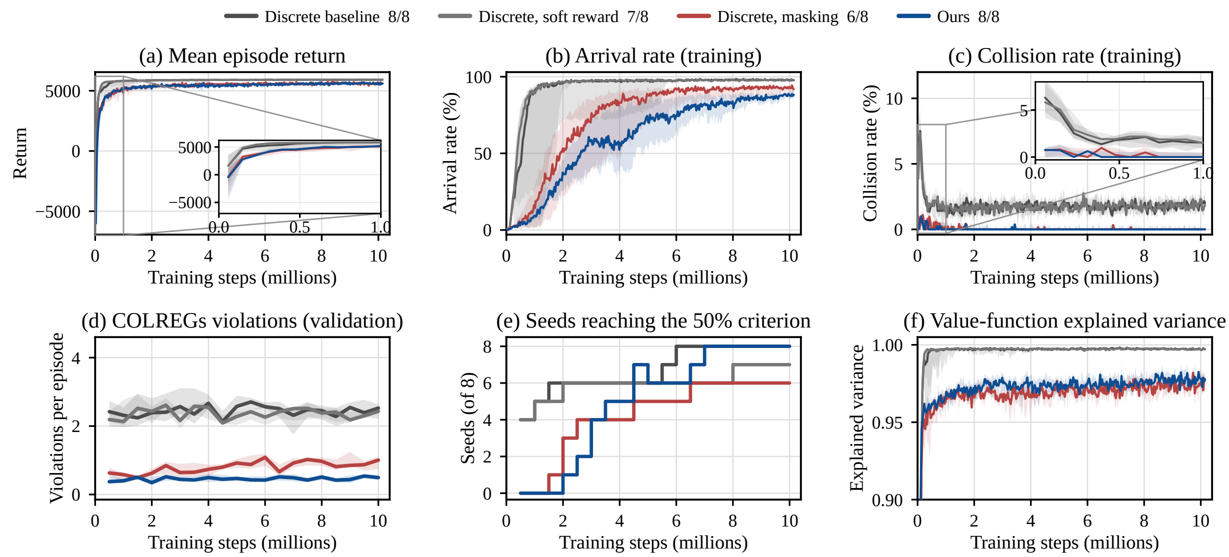

Figure 3 Training curves. Medians with interquartile bands over converged runs only, so the panels describe learning speed among those runs; the legend gives the number of converged runs out of eight. Panels (a) to (c) and (f) come from training rollouts, (d) is measured on the validation set once per checkpoint, and (e) counts all eight runs. The reward function differs across configurations, so the level in panel (a) is not compared across them.

bounded-only configuration, at 0.066. That ordering matches the turn increments in Figure 2.

The timings below come from one thread of an AMD EPYC 9654 processor. They cover 1,872 decision steps of the trained policy. The median solve time is 4.11 ms and the 99th percentile is 8.85 ms. The forward pass of the policy adds a median of 0.44 ms. A whole decision step, including the environment, has a median of 5.30 ms and a 99th percentile of 10.14 ms. The decision period is 10 s, so the 99th percentile of a step is about one thousandth of the period. These wall-clock times come from one machine and indicate an order of magnitude rather than a comparison against another method.

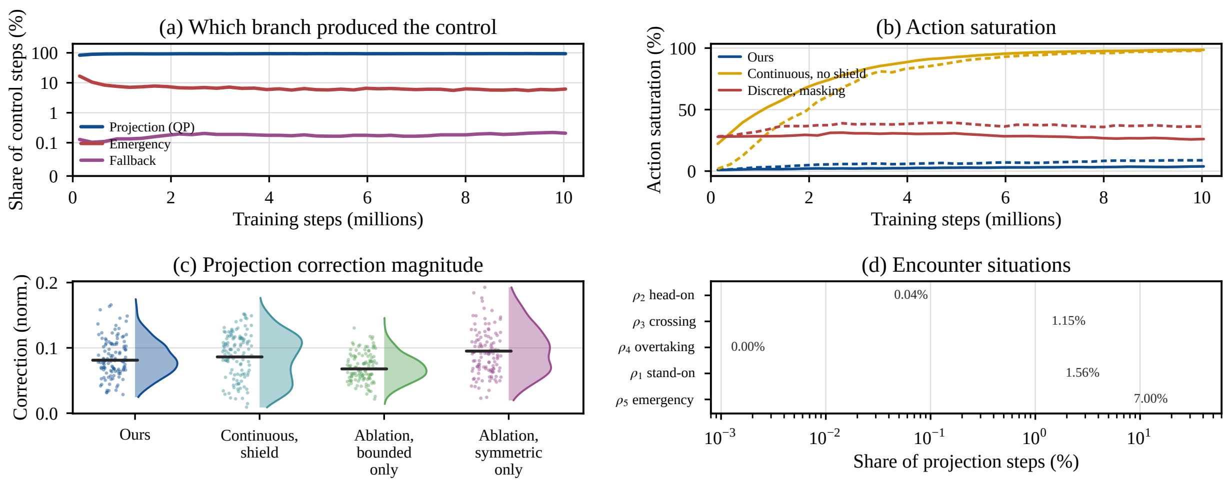

Figure 4 Behavior of the shield. Panel (d) is the share of projection steps in each situation, on a logarithmic scale. This figure is collected during training rather than at evaluation.

5.7 Comparison with Geometric Baselines

Three geometric collision-avoidance controllers that need no training are run through the same evaluation pipeline, on the same test set, at the same sample size (Table 4).

The comparison narrows the defensible claim to one dimension. The collision rate does not separate the two families: the barrier function baseline also reaches 0.00% here. The arrival rate and the smoothness columns are governed by the terminal constraint and by the control range, so neither is attributable to the shield. The largest observed difference is in COLREGs violations, and almost all of it comes from stand-on violations. These three controllers impose collision-avoidance constraints but no course-and-speed constraint, so they keep maneuvering while the rules require course and speed to be held. Ours records 0.53 violations per episode against 4.49 for the control barrier function at its lowest-violation setting. Almost all of that gap is in stand-on violations, at 0.36 against 4.04. Figure 5 shows tracks from two encounters of the test set, one head-on and one crossing. Three seeds were declared in advance for trajectory recording, and s_0 is the first of them. Both are the tightest constant-velocity approaches of their type. The three configurations differ mainly in how much course they give up to keep clear. The unshielded discrete baseline stays within 21 and 28 degrees of its initial heading and passes at 613 and 718 m. Ours turns by up to 53 and 64 degrees and passes at 1,150 and 889 m.

TABLE 4 Geometric baselines. Official test set, $N = 600$.

Method	Control range	Arrival %	Coll. %	COLREGs violations / episode			Turn Δ	Jerk
				Total	Give-way	Stand-on		
Control barrier function	Policy range, margin	69.83	0.00	4.49	0.45	4.04	0.00235	0.167
PD controller	Actuator limits	57.67	16.67	8.10	0.63	7.47	0.00032	0.040
Velocity obstacle	Actuator limits	42.67	0.00	8.06	0.68	7.38	0.00128	0.071
Ours	Policy range	(see Table 3)						

Notes. These controllers need no training and have no seed. Each is reported at the setting with the fewest violations per episode, tuned on a set disjoint from the one used here. Each was also run at the other control range, which leaves the ordering of the three methods unchanged. The policy range is ± 0.048 m/s^2 and ± 0.018 rad/s, which matches the output range of the trained policies, and the actuator limits are ± 0.24 m/s^2 and ± 0.03 rad/s. For Ours the shield and the emergency controller may use the actuator limits, so the executed range is not identical. Both are reported so that a difference in authority is not a confounder. The arrival rate is not a measure of collision avoidance. These controllers have no terminal alignment stage. Across the seven settings in this table they reach the goal position in 96.8% to 99.7% of episodes and then fail the terminal heading gate.

6 DISCUSSION

6.1 Continuous-Action Refinements and Guarantee Scope

A quantized action space has no samples pushed past the action box, and no distribution whose differential entropy is unbounded above. The bounded action distribution therefore addresses an issue specific to continuous actions. Symmetric entry addresses the state machine instead, and the same asymmetry can arise on a discrete action space. The ablation evaluates two implementation effects rather than two hyperparameters. Feasibility at later steps is not claimed. No proof is given that a feasible control command remains available at every subsequent decision step under a 10 s zero-order hold. A terminal constraint on the fallback maneuver is one route to that property. The closed-loop integration is not finished, so it is stated as a design direction and not as a guarantee. The construction does give an object that can be checked step by step: the classification of the situation, the direction of the half-plane, and the feasibility of the projection. None of the three depends on the training outcome.

6.2 Projection Versus Enumeration

The gain is in the resolution of a compliant turn. The rules require the heading change over the maneuver window to reach a threshold, which converts to a smallest admissible turn rate ω_{turn}. A continuous command reaches that value exactly. On a fixed grid the smallest available compliant turn rate is the first grid point at or beyond it, which coincides with the threshold only when the two divide. On the 7×7 grid reproduced here it is 37.5% above that value (Figure 6a).

That gap does not shrink monotonically as the grid is refined. The overshoot is set by the divisibility of ω_{turn} by the grid spacing, not by the spacing itself. A 5×5 grid overshoots by 3.1% while a 7×7 grid overshoots by 37.5%. A continuous command with projection reaches the smallest admissible turn rate for any threshold.

The cost is in computation. Enumerating and masking $K \times K$ commands scales with K^2, whereas the projection solves a fixed two-variable quadratic program. At the grid sizes used here enumeration is the faster of the two

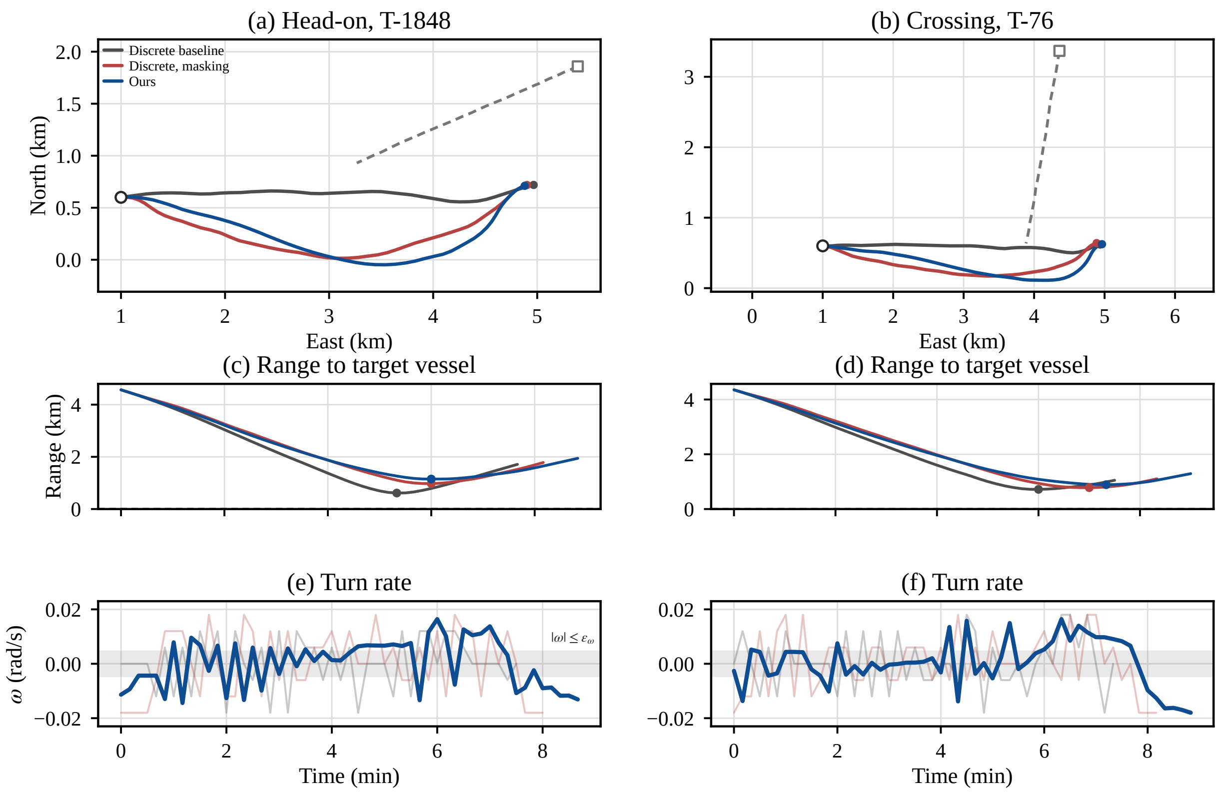

Figure 5 Two encounters, by column. Scenarios are picked by geometry alone: the smallest constant-velocity closest approach of each situation, 7 m for T-1848 and 2 m for T-76. Seed s_0, at the last checkpoint rather than the one selected on the validation set. Each column is one encounter, with its range and turn rate below the track. In (a) and (b) the circle and the square are the starts of the own and the target vessel, and the dashed line is the recorded target track. In (c) and (d) the dashed line is the closest approach under constant velocity. The shaded band in (e) and (f) is the stand-on tolerance of the shield; the offline scorer applies a different test (Section 5.2). In (e) and (f) the two discrete configurations are drawn faint so that the turn rate of Ours reads against them. Its mean step-to-step change in turn rate is 0.0079 and 0.0060 rad/s, against 0.0110 to 0.0131 for those two.

(Figure 6b). The case for projection is capability rather than speed: enumeration does not apply to a continuous set, while the projection lands exactly on the boundary of the constraint.

The difference in throttle resolution, where the discrete configurations have only seven levels, is structural in the same way. It cannot be read directly from the throttle increment column, because that column is also affected by the action-increment penalty, and that penalty is active only in the continuous configurations. The attributable comparison holds only among configurations that run the same penalty.

7 CONCLUSIONS

The give-way clauses of COLREGs require a vessel to turn to a stated side, so compliance cannot be verified from whether contact occurred. Under the action-space formulation adopted here, the implemented directional give-way requirement is a half-plane constraint on the control command. The premise that the safe control set must be enumerated can therefore be replaced by the premise that it is convex. That step applies a projection-based safety shield to maritime COLREGs under continuous control. Directional compliance on every give-way step where the projection is feasible holds by construction, independently of the training outcome and of the seed.

On the official test set of 600 scenarios the median violation count of Ours is 0.53 per episode. The discrete shielded baseline records 1.16 and the geometric baseline with the fewest violations per episode records 4.49. It is the only shielded configuration that meets the convergence criterion on all eight seeds.

The guarantees are scoped, and Sections 4 and 5 state each scope where it arises. The emergency and fallback branches bypass the constraint by design, so the measured violation count is not zero, and freedom from collision is

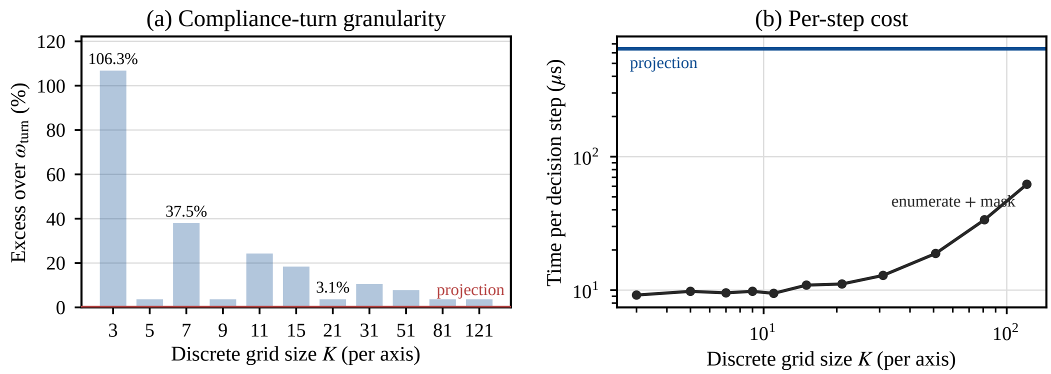

Figure 6 Projection against enumeration. (a) The amount by which the smallest compliant turn rate available on a $K \times K$ grid exceeds the smallest admissible value ω_{turn}. It is set by divisibility and does not fall monotonically with K. A continuous command with projection reaches it exactly, shown by the red line. (b) Time for one decision step. Enumeration grows with K^2, while projection does not depend on K. Panel (a) is exact arithmetic. Panel (b) is measured on one machine and is used only for the growth in K, not as the deployment cost.

not guaranteed. Propositions 1 and 3 assume a single target vessel at constant velocity, and Proposition 2 assumes a correct decision by the state machine. The overtaking branch remains untested because the scenario library holds no overtaking case, and wind, current, and waves are outside the vessel model.

Three directions follow. The terminal constraint of the fallback maneuver should be wired into the deployed shield and verified in closed loop. Directional compliance should be extended to several target vessels. The far-field test should become a constant-time early exit.

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AUTHOR CONTRIBUTIONS

Study conception and design: all authors. Method and software, and draft preparation: the first author. Data collection and analysis: the first and second authors. Interpretation of results: all authors. All authors reviewed the results and approved the final version.

USE OF GENERATIVE ARTIFICIAL INTELLIGENCE

The authors used Claude (Anthropic) and ChatGPT (OpenAI) while preparing this manuscript. The tools edited the English text from a draft written by the authors and checked the manuscript against the submission requirements. The study design, the software, the experiments, and the interpretation are the work of the authors, who reviewed every AI-assisted output and take full responsibility for this paper.

CONFLICT OF INTEREST

The authors declare no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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